



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

Water

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	120.0	100.0	100.0	°C
p1	☒ 90.0	90.0	90.0	MPa(g)
☐ p2	☒ 80.0	80.0	80.0	MPa(g)
☒ Δp	10.0	10.0	10.0	MPa
Cv	4.6153	4.6153	4.6153	GPM(US)
☐ qm	39,843.0	39,843.0	39,843.0	kg/h
☒ qv	40.0	40.0	40.0	m³/h
LpAe	64.8	64.8	64.8	dB(A)
	<i>Non-critical</i>	<i>Non-critical</i>	<i>Non-critical</i>	

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	996.07	996.07	996.07	kg/m³
ρ2	996.07	996.07	996.07	kg/m³
☒ η1	0.306	0.306	0.306	mPa s
☐ ν1	0.30721	0.30721	0.30721	mm²/s
pv1	1.9867	1.0142	1.0142	bar(a)
pv2	2.1025	1.0841	1.0841	bar(a)
tv1	373.95	373.95	373.95	°C
cF1	1,704.9	1,716.4	1,716.4	m/s

Pipe downstream of valve

Size class downstream of valve

NPS2

2"

Schedule downstream

SCH2

STD



Calculation Sheet

Valve configuration

Valve type	<i>Straight globe valve</i>
Trim type	<i>Contoured plug</i>
Flow direction	<i>FTC</i>
Valve performance class	<i>Multi stage valve</i>
Protection	<i>Non-hardened</i>
Low-noise design	<i>n.a.</i>

Valve data

Basic characteristic		<i>Linear</i>	
☐ Lift stopper			
Selected valve size	NPS	<i>2"</i>	
Pressure class	class ☐	<i>class 600</i>	
Maximum operating pressure	p,max ☐	<i>105.0</i>	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	<i>0.962</i>	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	<i>0.944</i>	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	<i>0.703</i>	-
Valve style modifier (Cv/Cv100 = 1)	Fd	<i>0.41</i>	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
u2	<i>5.482</i>	<i>5.482</i>	<i>5.482</i>	m/s
LpAe	<i>64.8</i>	<i>64.8</i>	<i>64.8</i>	dB(A)
Ri	<i>0.09</i>	<i>0.09</i>	<i>0.09</i>	-

Noise calculation

Calculation standard (noise, liquid)	<i>IEC 60534-8-4 (2015)</i>
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Noise prediction data

Low-noise design		<i>n.a.</i>	
Sound velocity	cF1	<i>1,704.9</i>	m/s
Pressure level	LA1	<i>177.0</i>	dB
Sound pressure level	Lpi	<i>177.0</i>	dB
Transmission loss	TL	<i>-95.9</i>	dB



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Noise prediction data (continued)

Sound power level (A-weighted)	LwAe	78.3	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	64.8	dB(A)

Comments:**Reliability index - Ri**

Seat tightness (Δp): 100 %

Recommended solution: Heavy duty valve, Hardened seat/plug

Reliability index - Ri,op2

Seat tightness (Δp): 100 %

Recommended solution: Heavy duty valve, Hardened seat/plug

Reliability index - Ri,op3

Seat tightness (Δp): 100 %

Recommended solution: Heavy duty valve, Hardened seat/plug

Legend

 Errors